

Performance Assessment of New Dual-Pocket Vertical Heterostructure Tunnel FET-Based Biosensor Considering Steric Hindrance Issue

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Abstract—In this paper, a new dual-pocket (DP), dielectric modulated (DM) heterostructured tunnel field-effect transistor (DM-HTFET)-based biosensor has been reported. First, the efficacy of the DP-hetero-TFET (HTFET) has been analyzed by comparing it with other existing TFET structures. Next, a comprehensive assessment of sensitivity between single-pocket (SP) and DP-DM-HTFET biosensors for pocket thickness (T_{pocket}), pocket length (L_{pocket}), pocket doping (N_{pocket}), work function of the gate metal, molar fraction of Ge, gate oxide layer, and gate oxide layer thickness (T_{ox}) was done. Hence, a nonuniform arrangement of biomolecules inside the cavity simulation has been done using ATLAS device simulation software to validate the working ability of the proposed sensor. Significant improvement in the sensitivity due to threshold voltage (on-current) i.e., 26.78% (78.5%), 60.8% (40.4%), 56% ($2.2 \times 10^2\%$), and 40.6% (80.68%) has been observed for the DP-DM-HTFET over SP with the variation of T_{pocket} , L_{pocket} , N_{pocket} , and T_{ox} , respectively. DP-DM-HTFET-based current mirror circuit has also been demonstrated at the end. Sensitivity evaluation discloses that the DP-DM-HTFET can be a promising candidate for CMOS-based label-free biosensing applications.

Index Terms—Dielectric modulated (DM) tunnel field-effect transistor, heterostructure, label-free biosensing, line and point tunneling, sensitivity.

I. INTRODUCTION

DIELECTRIC modulated (DM) field-effect transistor (FET)-based biosensors have drawn enormous consideration for label-free sensing of neutral as well as charged bioanalytes. The operative capacitance of the gate oxide layer changes with the dielectric constant (k) and the charge density (N_f) of bioanalytes inside the cavity region [1], [2]. Conversely, a conventional FET-based

biosensor [3] faces severe drawbacks like greater subthreshold swing and achieving a minimum recognition time caused by limitations of the KT/q factor [4]. To surmount these drawbacks, a tunnel FET (TFET)-based biosensor [5], [6] has been developed with better sensitivity, low dependence on the fill-factor, and faster response time due to band-to-band tunneling (BTBT) [7]. Short-gate (SG) formation in DM-heterostructured TFET (HTFET) architecture [8] attains superior sensitivity in assessment with a full-gate (FG) structure. Moreover, SG with dual-metal configuration, having an underlap model similar to a dual-metal-lateral DM-HTFET (L-DM-HTFET) arrangement offers improved sensitivity in terms of subthreshold slope (SS) and variation in on-current (I_{ON}) [9] than its FG counterpart. Here, the subthreshold (V_T) regime has been chosen as the sensitivity becomes optimum [10] in the sub- V_T regime. The recognition of bioanalytes is mainly dependent on energy band (EB) bending at the tunneling interface [5]; hence, the location dependence of the sensitivity parameter can impose severe restrictions on the TFET because of the low binding probability of bioanalytes inside the cavity zone [6]. To overcome this drawback, vertical (line tunneling) and lateral tunneling (point tunneling) can be recognized by using a pocket TFET concept where the pocket is doped dissimilar to that of the source. This pocket TFET arrangement offers less sensitivity over the position of bioanalytes inside the cavity since both line and point tunneling elements supply current [3].

To enhance I_{ON} and steeper SS, a hetero-TFET [11] is preferable. Although single pocket (SP)-HTFET [12]–[14] biosensor has been reported in the literature, dual-pocket (DP) HTFET (DP-HTFET) arrangement has not been used for biosensing applications till date. Hence, back gate and front gate n^+ -pockets have been introduced in such a way that they can exist on the bottom and top surfaces of the device, in the source region, close to the source-channel interface. As overlapping gates in the source zone offer more electrostatic influence over the channel area, I_{ON} is improved further. Hence, to verify the efficacy of the DP-HTFET, first a comparative analysis has been performed with the lateral (L) and SP-HTFET in terms of characteristic parameters, as given in Table I. Considering the L-HTFET structure as a reference, improvement of different parameters for the SP-HTFET and the DP-HTFET has been detailed in Table I. The DP-HTFET outperforms due to the occurrence of lateral and vertical tunneling events and also because of DP architecture. This

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TABLE I
PERFORMANCE ASSESSMENT OF SHORT-GATE L, SP, AND
DP-HTFET WITHOUT CAVITY STRUCTURE

Parameters	$V_{gs}=1.3$ V, $V_{ds}=0.8$ V, $x=0.5$, $\Phi_{M1}=4.3$ eV, $\Phi_{M2}=3.8$ eV					
	L-HTFET		SP-HTFET		DP-HTFET	
	Value	ΔP (%)	Value	ΔP (%)	Value	ΔP (%)
DIBL (mV/V)	125.3	0	85.7	-31.6	60.2	-51.95
SS (mV/dec.)	89.6	0	64.3	-28.2	48.4	-45.98
I_{ON} ($\mu A/\mu m$)	0.56	0	5.3	834	19.3	3321
V_{TH} (V)	0.62	0	0.4	-31.2	0.2	-71.36
E_{max} (MV/cm)	2.39	0	2.4	2.92	2.6	10.87
TR_{max} (cm ² S)	28.6	0	29.1	1.7	29.7	3.84
$\Delta P(\%)$ denotes % of improvement which is measured with respect to L-HTFET structure						

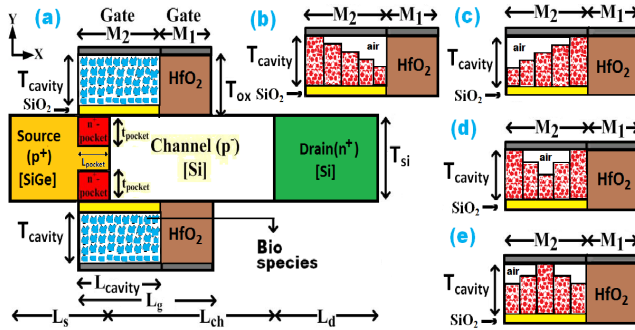


Fig. 1. Schematic of dual-metal short-gate DP-DM-HTFET having (a) fully filled cavity, (b) decreasing step patterns, (c) increasing step patterns, (d) concave step patterns, and (e) convex step patterns of bioanalytes.

DP-HTFET has been reshaped by etching a certain portion of the oxide layer to form DM DP-HTFET (DP-DM-HTFET) for the first time in the literature, to the best of our knowledge.

The majority of researchers have investigated completely filled cavity [5], [8], [15], [16]. Hence, in the proposed architecture, the moderately filled cavity has been considered as the array of bioanalytes inside the cavity is uneven and composite due to low binding probability, the functionality of the surface [17], [18], and so on. This approach would be more practical. Besides, the steric hindrance issue does not allocate the cavity to be entirely filled [19]–[21]. Hence, various nonuniform step patterns such as decreasing, increasing, concave, and convex are considered inside the cavity as shown in Fig. 1(b)–(e).

II. PROPOSED DP-DM-HTFET BIOSENSOR STRUCTURE

A 2-D schematic of DP-DM-HTFET biosensor architecture incorporating regular/irregular arrangements of biomolecules has been proposed by following the International Technology Roadmap for Semiconductors [22] and is depicted in Fig. 1. The proposed TFET structure with a double-gate configuration is utilized to easily access the well-setup device-stage simulation frame [23]. The DP-DM-HTFET biosensor has the channel length (L_{ch}) = 42 nm; source/drain length ($L_{S/D}$) = 20 nm; cavity length (L_{cavity}) = 15 nm; silicon film thickness (T_{Si}) = 10 nm; HfO₂oxide layer thickness (T_{ox}) = 6 nm, and nanogap cavity thickness (T_{cavity}) = 5 nm [8], [14].

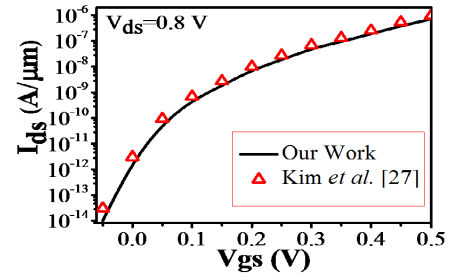


Fig. 2. Proposed model calibration (I_{ds} – V_{gs} curves) against the experimental records in [27].

The process flow to form a nanogap cavity by etching a certain part of gate oxide for DP-DMTFET is illustrated in [18]. The uniform doping concentrations of the source, channel, and drain regions are $5 \times 10^{19}/cm^3$, $1 \times 10^{12}/cm^3$ and $5 \times 10^{18}/cm^3$, respectively. Hence, $N_{pocket} = 5 \times 10^{19}/cm^3$, $T_{pocket} = 3$ nm, and $L_{pocket} = 5$ for the n^+ -pocket regions. In the proposed structure, low bandgap ($Si_{1-x}Ge_x$) with Ge molar fraction (x) = 0.5 is chosen as a source region to obtain better I_{ON} [24]. HfO₂ is used as an oxide layer, and tin ($\Phi_{M1} = 4.3$ eV) on the source side and aluminum ($\Phi_{M2} = 3.8$ eV) on the drain side have been used as dual-gate metals. For the lateral structure, gates M_1 and M_2 are 10 nm each for both the front gate and the back gate, while for SP it is 10 and 15 nm for the front gate and 10 nm each for the back gate. Here, for the DP structure, gates M_1 and M_2 are of 10 and 15 nm each for both the front gate and the back gate, respectively. As $\Phi_{M1} > \Phi_{M2}$, little modification in V_{ds} can proficiently reduce the hot carrier effect [25] and improve the longevity of the device. Also, better control of the gate over the channel improves the carrier transport effectiveness.

Silvaco ATLAS, a commercially available numerical simulator [26], has been used to simulate the device structures. An extremely popular nonlocal BTBT model is preferred to compute generation probability at the tunneling interface. Besides, the Shockley-Read-Hall (SRH) recombination/generation model, the Auger model, the Fermi–Dirac carrier distribution, the bandgap narrowing (BGN) model, the electric field-dependent mobility (FLDMOB) model, and the quantum confinement model to set quantum tunneling are taken into consideration during the simulation. To validate and standardize these models and methods, simulation results are calibrated against Kim *et al.* [27], which is illustrated in Fig. 2. The calibration is performed for a model with a Ge source, since experimental records on the SiGe TFET related to line (vertical) tunneling are not accessible. Fig. 2 provides an explanation for such a calibrated model, for which the device architecture, together with its dimensions and design parameters, has maintained the same, as utilized in [27]. Default material parameters concerned with Ge, as offered in Silvaco ATLAS, are exploited for the source zone to initiate the simulated I_{ds} – V_{gs} curves. An excellent agreement between the outcomes of the two works has been observed in Fig. 2.

In this configuration, we have included a layer of SiO₂ with 1 nm thickness to comprise the growth of native oxide within the cavity once the silicon substrate is exposed to the air environment [28], [29]. Here, the SiO₂ layer, the in-between cavity, and the silicon body prevent gate-channel leakage and also serve as an adhesive layer to immobilize the

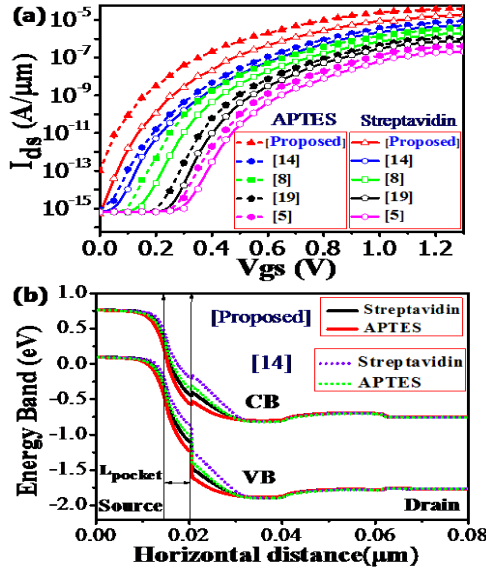


Fig. 3. Assessment of (a) I_{ds} - V_{gs} characteristics of the proposed structure with other existing literatures [5], [8], [14], [19] at $V_{ds} = 0.8$ V, $N_f = 0$. (b) EB diagram of SP and DP DM-HTFET along horizontal distance under ON-condition in presence of neutral bioanalytes ($N_f = 0$) with regular arrangement.

bioanalytes [28], [29]. APTES (3-aminopropyltriethoxysilane, $k = 3.57$) and biotin ($k = 2.63$) can be used as the linker and bioreceptor, respectively, to detain streptavidin ($k = 2.1$) as the biotarget [15], while $k = 1$ denotes that the cavity is filled up with air. Conversely, besides the dielectric constant, characteristics of charged bioanalytes, e.g., nonhybridized single-strand DNA, have the uniqueness both in terms of the charge and the dielectric constant [30]. Accordingly, the presence of charged bioanalytes inside the cavity is imitated in the simulation by including oxide charges ($N_f = \pm 5 \times 10^{15}/m^2$, $\pm 1 \times 10^{16}/m^2$) at some appropriate location surrounded by the gate insulator itself.

III. RESULTS AND DISCUSSION

A. Performance Assessment of the Proposed Biosensor Considering the Regular Arrangement of Bioanalytes

Fig. 3(a) illustrates the I_{D} - V_{GS} and the EB diagram of the proposed biosensor with existing structures, i.e., SP-DM-HTFET [14], L-DM-HTFET [8], SP-DM-TFET [19], and the conventional TFET [5] in terms of transfer characteristics when the linker (APTES) and the bioreceptor (biotin) are present to detain the biotarget (streptavidin) inside the cavity with a regular arrangement at $V_{ds} = 0.8$ V and $N_f = 0$, respectively. Fig. 3(b) demonstrates the EB diagram of SP and DP DM-HTFET under the abovementioned conditions ($V_{ds} = 0.8$ V, $V_{gs} = 1.3$ V). With the increment of k , due to the higher capacitive effect, the band bends significantly at the source/pocket interface for SP and DP DM-HTFET. Consequently, V_{TH} reduces and I_{ON} increases. Moreover, the occurrence of lateral and vertical tunneling events and the presence of dual-gate n^+ -pockets improve the BTBT in the proposed structure. Hence, DP-DM-HTFET offers better I_{ds} than SP counterpart.

Hence, 25%-, 50%-, and 100%-filled cavities are expressed as quarter filled (QF), half-filled (HF), and full filled (FF),

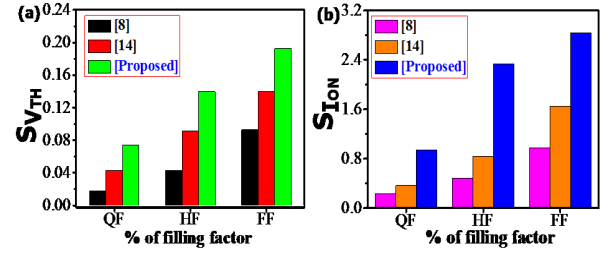


Fig. 4. Assessment of sensitivity parameter (a) $S_{V_{TH}}$ and (b) $S_{I_{ON}}$ between DP-DM-HTFET [proposed], SP-DM-HTFET [14] and L-DM-HTFET [8] with respect to percentage of filling factor at $V_{ds} = 0.8$ V, $N_f = 0$ with a regular arrangement of streptavidin.

respectively. Streptavidin is considered as the biotarget here. Sensitivity analysis ($S_{V_{TH}}$ and $S_{I_{ON}}$) of L, SP, and DP DM-HTFET is detailed in Fig. 4(a) and (b) at $V_{ds} = 0.8$ V. The sensitivity factors $S_{V_{TH}}$ and $S_{I_{ON}}$ can be expressed as $S_{V_{TH}} = (V_{TH}^{air} - V_{TH}^{bio})/V_{TH}^{air}$ and $S_{I_{ON}} = (I_{ON}^{air} - I_{ON}^{bio})/I_{ON}^{air}$, respectively, where V_{TH}^x (I_{ON}^x) indicates V_{TH} (I_{ON}) when the cavity is filled by x (= air or bioanalytes). Completely air-filled cavity is considered as a reference. Sensitivity parameters increase with the percentage of filling factor and the maximum is at the FF condition. $S_{V_{TH}}$ is improved by 107.1% (50.5%) for DP (SP) over [8] and $S_{I_{ON}}$ is improved by 191.2% (68.8%) for DP (SP) over [8] due to the better BTBT effect in the proposed structure.

B. Performance Assessment of the Proposed Biosensor Considering the Irregular Arrangement of Bioanalytes

1) **Sensitivity Assessment and Energy Band Analysis:** Due to the steric hindrance issue [19]–[21], various nonuniform step patterns of biomolecules have been considered inside the cavity. The evaluation of sensitivity for different patterns is shown in Fig. 5(a) and (c) when the linker (APTES) and the biotarget (streptavidin) are immobilized inside the cavity as neutral bioanalytes at charge density $N_f = 0$ and as charged bioanalytes having $N_f = \pm 1 \times 10^{16}/m^2$ when the biotarget (streptavidin) exists as neutral bioanalytes, respectively. With the increment of k of bioanalytes, electrostatic coupling between the gate and the channel enhances in the area of elevated steps nearer the tunneling interface. Hence, the sensitivity of the biosensor is superior in decreasing and concave step patterns with respect to the other arrangements as the elevated steps are positioned distant from the source-to-channel tunneling interface. It is apparent from Fig. 5(b) and (d), which illustrates the assessment of EB diagram between nonuniform step patterns along the horizontal distance under ON-condition ($V_{ds} = 0.8$ V, $V_{gs} = 1.3$ V) when the linker (APTES) is immobilized inside the cavity as a neutral bioanalyte at the charge density, $N_f = 0$ and charged bioanalytes having $N_f = +1 \times 10^{16}/m^2$ when the biotarget (streptavidin) exists as neutral bioanalytes, respectively. As the decreasing step pattern provides the best results, this pattern would be considered during the comparative analysis of SP- and DP-DM-HTFET for the remaining portion of this paper.

2) **Impact of Variation of T_{pocket} , L_{pocket} , and N_{pocket} :** Fig. 6(a)–(c) illustrates the assessment of $S_{V_{TH}}$ and $S_{I_{ON}}$ for variation of T_{pocket} , L_{pocket} , and N_{pocket} at $V_{ds} = 0.8$ V. As T_{pocket} enhances from 2 to 5 nm, the tunneling width reduces and the electron tunneling from the source to the

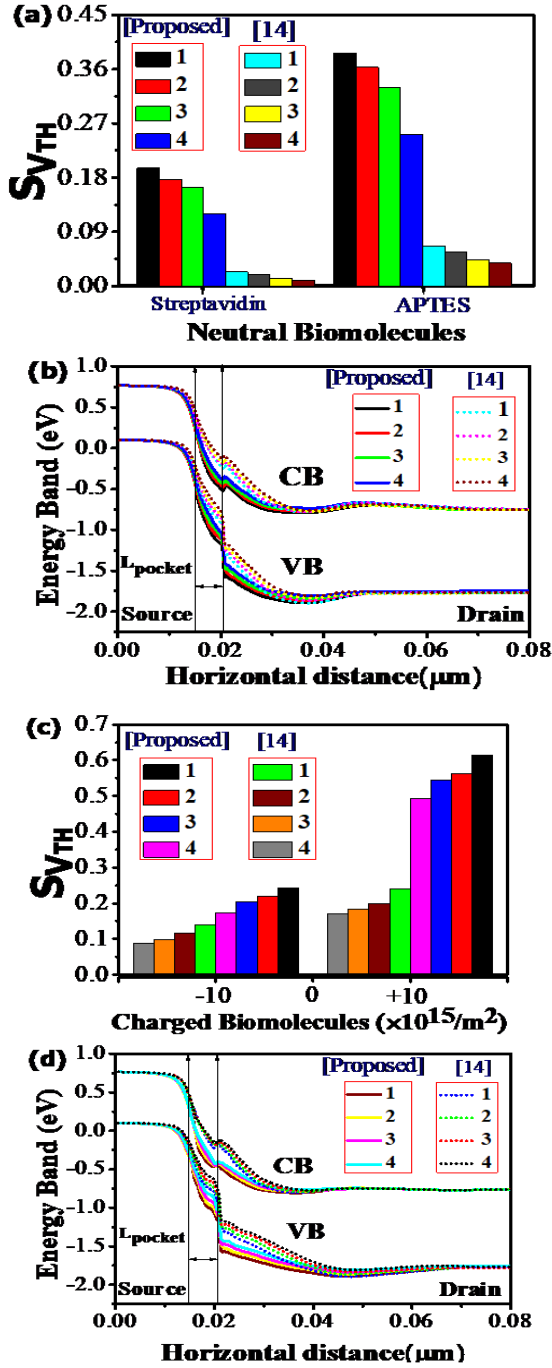


Fig. 5. Assessment of (a) and (c) sensitivity and (b) and (d) EB diagram of nonuniform step patterns for SP- and DP-DM-HTFET in the presence of neutral bioanalytes ($N_f = 0$) and charged bioanalytes (streptavidin as a biotarget) at $V_{ds} = 0.8$ V. Here, 1: decreasing, 2: concave, 3: convex, and 4: increasing patterns of biomolecules.

channel region increases. Consequently, V_{TH} reduces and I_{ON} increases. Hence, in the proposed biosensor, $S_{VTH}(S_{ION})$ is improved by 26.78% (78.5%) compared to [14] for $T_{pocket} = 5$ nm due to advanced BTBT incident. The higher value of T_{pocket} is favored to achieve superior performance of the proposed biosensor. However, for $T_{pocket} > 5$ nm, V_{TH} increases and I_{ON} gradually reduces due to less band-bending near the source-channel interface. Hence, the maximum T_{pocket} is 5 nm. When L_{pocket} enhances, then the source to the channel tunneling width reduces, and consequently, V_{TH} (I_{ON}) reduces

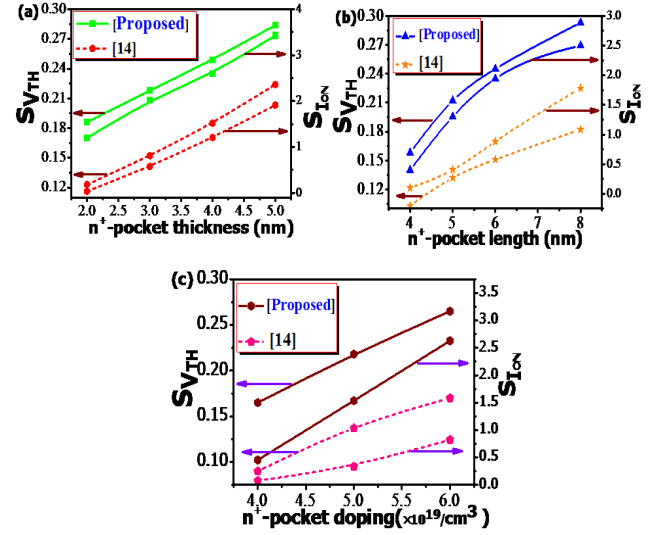


Fig. 6. Evaluation of sensitivity between the DP-DM-HTFET [proposed] and the SP-DM-HTFET [14] with variation of (a) T_{pocket} (b) L_{pocket} , and (c) N_{pocket} at $V_{ds} = 0.8$ V, $N_f = 0$ in the presence of streptavidin as a biotarget.

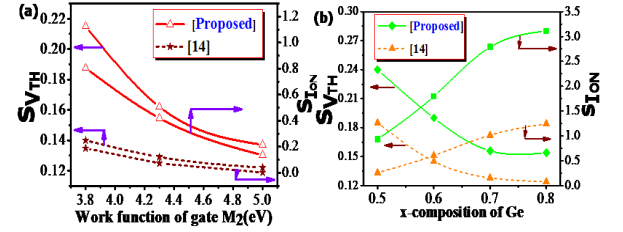


Fig. 7. Assessment of sensitivity between the DP-DM-HTFET [proposed] and the SP-DM-HTFET [14] with variation of (a) Φ_{M2} and (b) molar fraction of Ge at $V_{ds} = 0.8$ V, $N_f = 0$ in the presence of streptavidin as a biotarget.

(increases). Hence, $S_{VTH}(S_{ION})$ is improved by 60.8% (40.4%) for the DP-DM-HTFET over [14] for $L_{pocket} = 8$ nm due to the presence of a dual n^+ -pocket in the configuration. A large value of L_{pocket} is required to realize the better performance of the proposed biosensor.

Hence, $T_{pocket} = 3$ nm and $L_{pocket} = 5$ nm are kept the same as [14] for comparative analysis.

With the increase in N_{pocket} from $4 \times 10^{19}/\text{cm}^3$ to $6 \times 10^{19}/\text{cm}^3$, the current density within the channel beneath the gate region enhances due to the narrow source-to-channel tunneling width [31]. Hence, $V_{TH}(I_{ON})$ decreases (improves). The maximum improvement of 56% (220%) on $S_{VTH}(S_{ION})$ has been observed in the proposed model over [14] at $N_{pocket} = 6 \times 10^{19}/\text{cm}^3$. This is mainly due to improved BTBT effect. Hence, higher N_{pocket} is essential to achieve better efficacy of the proposed biosensor. $N_{pocket} = 5 \times 10^{19}/\text{cm}^3$ is kept the same as [14] for comparative analysis throughout our work.

3) Sensitivity Comparison for Work Function Variation of Gate Metal M_2 : Fig. 7(a) depicts the comparison of S_{VTH} and S_{ION} due to variation in the work function (Φ_M) of gate metal M_2 . The high value of Φ_{M1} and the low value of Φ_{M2} in DP-DM-HTFET cause a steep increase in transfer characteristics, as shown in Fig. 7(a). When $\Phi_{M2} < \Phi_{M1}$, the tunneling width reduces and the ability of gate controlling at the source/channel junction enhances which improves I_{ON} . Here, $S_{VTH}(S_{ION})$ achieves maximum sensitivity degradation by an amount of 53.5% (327%) for DP-DM-HTFET over [14]

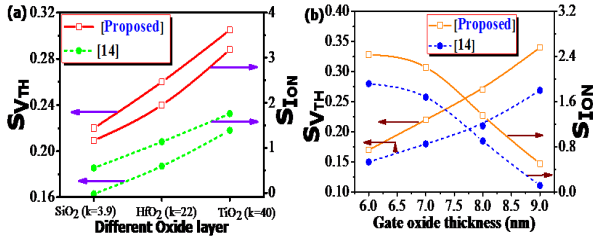


Fig. 8. Evaluation of sensitivity between the DP-DM-HTFET [proposed] and the SP-DM-HTFET [14] with variation of (a) different gate oxide layers (b) T_{ox} at $V_{ds} = 0.8$ V, $N_f = 0$ in the presence of streptavidin as a biotarget.

when $\Phi_{M2} = 3.8$ eV due to the presence of the dual n^+ -pocket in the configuration. Hence, $\Phi_{M2} = 3.8$ eV is considered as an optimized parameter.

4) *Impact of Variation of Molar Fraction of Ge:* Fig. 7(b) shows the effect of change in the molar fraction of Ge ($Si_{1-x}Ge_x$) [24] on $S_{V_{TH}}$ and $S_{I_{ON}}$, respectively. As the x -composition of Ge increases from 0.5 to 0.8, the peak tunneling current density (J_{peak}) enhances [21] due to the decrement of bandgap energy (E_g) as follows: $E_g = (1.17 - 0.896x + 0.396x^2 - 0.05)$ eV, x = molar fraction of Ge in the $Si_{1-x}Ge_x$ alloy. Hence, $S_{V_{TH}}$ ($S_{I_{ON}}$) gradually decreases (increases). $S_{V_{TH}}$ ($S_{I_{ON}}$) is degraded (improved) by a maximum amount of 29.7% (175%) for the DP-DM-HTFET [14] because of the BTBT effect. The Ge molar fraction of 0.5 in the $Si_{1-x}Ge_x$ alloy [24] has been set throughout the simulation.

5) *Sensitivity Assessment of Variation of Gate Oxide Layer:* Fig. 8(a) illustrates the effect of different gate oxides (SiO_2 , HfO_2 , and TiO_2) on $S_{V_{TH}}$ and $S_{I_{ON}}$. As the dielectric constant of the oxide layer (k) increases, the oxide capacitance ($C_{ox} = k\Phi_{ox}/T_{ox}$) enhances. Hence, V_{TH} (I_{ON}) will reduce (increase). Fig. 8(a) depicts that both $S_{V_{TH}}$ and $S_{I_{ON}}$ will rise up with the increase in dielectric constant in both structures. Considering TiO_2 as the gate oxide layer, $S_{V_{TH}}$ ($S_{I_{ON}}$) will be improved by 40.6% (80.68%) for the DP-DM-HTFET over [14] due to the presence of the dual n^+ -pocket in the proposed structure. Hence, gate oxide with a higher k -value is essential to achieve superior performance of the DP-DM-HTFET configured biosensor. During simulation, HfO_2 ($k = 21$) is retained as in [14] for relative analysis throughout this paper.

6) *Impact of Variation of Gate Oxide Thickness:* Fig. 8(b) demonstrates the effects of T_{ox} variations on $S_{V_{TH}}$ and $S_{I_{ON}}$. As T_{ox} increases, the impact of localized interface charges (N_f) on the threshold potential becomes more significant [32]. This is due to the reduction of C_{ox} , hence the flat band potential (V_{fb}) also increases, which is expressed by $V_{fb} = V_{fb0} + qN_f/C_{ox}$, where V_{fb0} signifies the flat band potential excluding N_f . Hence, 1×10^7 A/ μm current is achieved later, and so, I_{ON} gets reduced [31]. It can be noted that $S_{V_{TH}}$ ($S_{I_{ON}}$) attains maximum sensitivity improvement (degradation) by the amount of 26.39% (50.4%) for the DP-DM-HTFET over [14] when $T_{ox} = 9$ nm. Hence, the higher value of T_{ox} is favored to achieve superior performance of the DP-DM-HTFET configured biosensor located close to the tunneling interface. In order to get optimum performance, $T_{ox} = 9$ nm is considered throughout our work.

7) *Comparative Performance Analysis Between the DP-DM-HTFET and the Accumulation Mode FET for Biosensing Applications:* In this section, we focus on the comparative

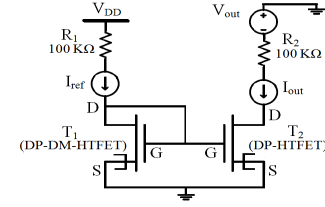


Fig. 9. Proposed DP-DM-HTFET-based current mirror circuit.

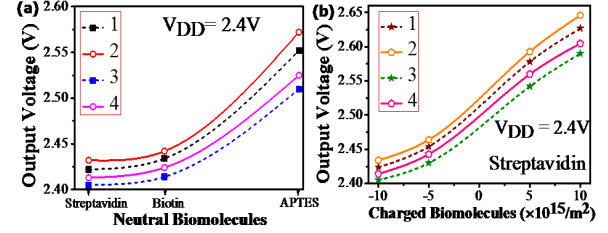


Fig. 10. Assessment of output voltage variation of the current mirror circuit in the presence of (a) neutral bioanalytes and (b) charged bioanalytes at $V_{DD} = 2.4$ V. Here, 1(2): irregular (regular) arrangement in the DP-DM-HTFET and 3(4): irregular (regular) arrangement in the SP-DM-HTFET based current mirror circuit.

performance analysis between the DP-DM-HTFET and the accumulation mode FET (AMFET) [33] in terms of $S_{V_{TH}}$ and $S_{I_{ON}}$ for decreasing step pattern of streptavidin as the biotarget. Although the TFET reveals superior sensitivity, the location dependence of the sensitivity parameter can impose severe restrictions on the TFET [6]. Since a superior sensitivity value over a wide range of bioanalytes position is needed, an AMFET exhibits enhanced performance than the TFET based biosensor [6]. But in this context, as the bioanalytes are placed close to the source-channel interface, the TFET outperforms its JLT contender. Significant improvement in $S_{V_{TH}}$ ($S_{I_{ON}}$) i.e., 20.3% (64.8%), 51.47% (32.63%), 67.25% ($1.73 \times 10^{-2}\%$), and 29.54% (56.73%) has been achieved due to variation of the T_{pocket} , L_{pocket} , N_{pocket} , and gate oxide layer, respectively. Thus, our proposed sensor device can serve as a pretty good sensor device for biosensing applications.

8) *Analog Application of the Proposed DP-DM-HTFET:* Current mirror-based potentiostats for a three-electrode amperometric electrochemical sensor is given in [34]. Fig. 9 demonstrates the basic arrangement of a current mirror circuit consisting of a proposed DP-DM-HTFET, in which the load current (I_{out}) is adjusted by the controlling current (I_{ref}) passing through another branch of a network. Here, T_1 is our proposed DP-DM-HTFET and T_2 is the DP-HTFET device. Conjugation of bioanalytes inside a cavity with their regular and irregular arrangements having different k and N_f values affects the I_{ds} of the DP-DM-HTFET, and the circuit arrangement is such that it can further adjust the load current to the same level. Fig. 10(a) and (b) depicts the assessment of output voltage (V_{out}) variation between the DP-DM-HTFET- and SP-DM-HTFET-based current mirror arrangements in the presence of neutral biomolecules having different k values and charged bioanalytes with their corresponding N_f for the supply voltage $V_{DD} = 2.4$ V. It is obviously observed that V_{out} attains maximum improvement by an amount of 1.86% (1.67%) with the regular (irregular) arrangement of APTES, while that for $N_f = +1 \times 10^{16}/m^2$ is 1.57% (1.42%) for the DP-DM-HTFET as it offers a lesser barrier width in comparison with that corresponding to the SP counterpart due

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